WASHINGTON MBELEKI

SOFTWARE ENGINEER | DATA ENGINEER

Johannesburg, South Africa | washiembeleki2003@gmail.com | 084 496 9910 | GitHub: github.com/WashingtonMbeleki | LinkedIn: linkedin.com/in/washie

PROFILE

Software engineering graduate targeting the Software Engineer (Python) position at takealot.com. Brings practical experience in building backend services, API integrations, and data persistence layers. Direct technical alignment for this role includes api, apis, data, etl, python, rest. Proven track record in backend service architecture, API integration, and clean code practices, supported by hackathon achievements and peer tutoring experience at WeThinkCode_.

DATA ENGINEERING SKILLS

Languages & query	| Python, SQL, Java, JavaScript, TypeScript, Bash/Shell
Data & storage	| SQLite, Cloudflare D1, Cloudflare KV, Relational data storage, Persistence, Caching, ETL concepts
Backend & integration	| REST APIs, Flask, Node.js, Hono, Cloudflare Workers, Microservices, JWT, Third-party API integration
Workflow & engineering	| Workflow automation, TDD, JUnit 5, Object-oriented design, Git/GitHub, GitLab CI/CD, Linux, Docker, Maven, Agile

EXPERIENCE

Peer Tutor & Bootcamp Assistant | *WeThinkCode_* Oct 2025 - Present

- Facilitate hands-on sessions covering Python fundamentals, Java object-oriented design, and Git workflows for incoming cohorts.
- Conduct code reviews, help troubleshoot systems, and reinforce test-driven development and maintainable coding practices.

SELECTED PROJECTS

Automated Job Pipeline & Tailored PDF Resume Engine | *Personal Project*

- Engineered an automated lead pipeline scraping public job feeds using Python, BeautifulSoup, and backoff retry handling.
- Implemented TF-IDF cosine similarity scoring to match candidate profiles against incoming job descriptions.
- Built a Jinja2 & xhtml2pdf rendering engine to automatically generate job-tailored PDF CVs and managed workflow tracking via a Streamlit dashboard and SQLite.

MoCredit - AI Credit Scoring & Micro-Lending Mini App | *2nd Place, MTN MoMo FinTech Hackathon, 2026*

- Worked on a fintech application using a Node.js/Hono backend on Cloudflare Workers, Cloudflare D1 for persistent storage, and Cloudflare KV for score caching.
- Integrated MTN MoMo Collections (RequestToPay) and Disbursements APIs, with an n8n-orchestrated AI coaching workflow.

Thuso Health - Digital Health Passport & Smart Clinic Finder | *Top 3, G13 Health Challenge*

- Co-built an offline-first health passport and clinic finder with backend APIs using Hono/Cloudflare Workers, D1 records, R2 medical attachments, and JWT authentication.
- Developed data-informed clinic recommendations using travel-time, queue-congestion, and load-shedding/solar-power information; supported offline caching and sync-on-reconnect.

TSUMO Network - Backend Data Architecture

- Co-engineered Python-based REST API endpoints handling media-streaming metadata workflows and user request routes.
- Implemented server-side business logic to automate complex, usage-based revenue allocation and payout calculations.
- Integrated application layers with persistent relational database storage to ensure safe, validated transaction logging.

Code Clinic Booking System | *WeThinkCode_ course project*

- Built a booking application for scheduling and managing academic code clinic sessions using Python, Flask, and SQLite.

Robot World Simulation Platform

- Built a Java multiplayer client-server simulation using Maven, object-oriented design, test-driven development, Swing, and socket/network programming; refactored code in response to complexity warnings.

EDUCATION

Occupational Diploma in Software Engineering | WeThinkCode_ Oct 2025 - Expected Feb 2027

Johannesburg, South Africa | *Relevant coursework:* Software Engineering Principles, Data Structures & Algorithms, Object-Oriented Design, Concurrent Systems, CI/CD, and Database Management.

TRAINING & CERTIFICATIONS

- Data Engineering Basics for Everyone — edX, Aug 2026.
- Relational Database Management Systems — IBM, Sep 2026.
- Generative AI for Software Engineering — WeThinkCode_; prompt engineering, AI-assisted refactoring, test-suite generation, and LLM API integrations.

ROLE-FIT SUMMARY

> *Role-Fit Summary (Software Engineer (Python) @ takealot.com): Key matching strengths include practical competence in api, apis, data, etl, python, rest, strong object-oriented design, and test-driven development. Demonstrates immediate value for takealot.com's development pipeline with a continuous commitment to expanding ETL and database engineering capabilities.*

WASHINGTON MBELEKI

Strategic Brief: Candidate Positioning & Value Proposition

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POSITIONING

Software engineer targeting entry-level data engineering roles, with hands-on project evidence across relational storage, API integration, caching, and data-informed application features. MoCredit and Thuso Health show how I connect data and services to support application workflows. My next step is to extend this foundation into clearly demonstrated ingestion, transformation, data-quality checks, and repeatable pipeline operation.

TARGET ROLES & WORKING SETUP

- **Roles:** Entry-Level / Junior Data Engineer; Data Engineering Intern or Trainee; Graduate Data Engineer; Junior ETL Developer; backend roles with substantial SQL and data-processing work.
- **Environment:** A team with mentorship, code review, clear data-quality expectations, and opportunities to build and operate small pipelines.
- **Application Strategy:** Prioritize roles that value Python, SQL, backend fundamentals, and learning potential. Treat Spark, Airflow, dbt, and cloud data warehouses as learning targets unless separate hands-on evidence is available.

VALUE PROPOSITION

- **Data-Backed Application Workflows:** MoCredit combines D1 persistence, KV caching, financial API integrations, and workflow automation.
- **Operational Data Features:** Thuso Health recommendations use travel-time, queue-congestion, and energy-availability information, alongside offline caching and sync considerations.
- **Engineering & Backend Discipline:** TSUMO Network, Code Clinic Booking, and peer tutoring provide evidence of API/service work, Python/SQLite, debugging, testing, code review, and teamwork.

PROOF POINTS

Project	What It Demonstrates	Stack
Job Pipeline Engine	Scraping, TF-IDF semantic matching, SQLite tracking, and dynamic PDF generation.	`Python, BeautifulSoup, TF-IDF, SQLite, Streamlit, xhtml2pdf`
MoCredit	Persistent storage, caching, external API integration, workflow automation; 2nd place at MTN MoMo FinTech Hackathon 2026.	`Node.js, Hono, Workers, D1, KV, n8n`
Thuso Health	Data-informed recommendations, offline-first behavior, records and attachment storage.	`Hono, Workers, D1, R2, JWT`
TSUMO Network	Backend services, REST endpoints, revenue calculations, persistence integration.	`Python, APIs, persistent storage`
Code Clinic	Database-backed booking workflow.	`Python, Flask, SQLite`

STRATEGIC BRIEF: GAPS & DEVELOPMENT PLAN

KNOWN GAPS & HOW I AM CLOSING THEM

- **No standalone end-to-end ETL project yet:** Build a public-data pipeline that ingests a CSV or API source, validates and transforms records with Python, and loads them into SQLite or PostgreSQL.
- **SQL depth needs visible proof:** Add examples for joins, aggregations, CTEs, window functions, null handling, deduplication, data-quality checks, and query optimization.
- **Reliability and orchestration:** Add repeatable runs, logging, environment-based configuration, error handling, and safe reruns. Add scheduling after core pipeline is stable.

FIRST 90 DAYS (PROPOSED)

Period	Focus	Evidence to Produce
Days 1-30	Build a complete Python + SQL pipeline	Public CSV/API ingestion, validation/transformation, relational database loading, unit tests and sample outputs.
Days 31-60	Improve SQL and pipeline reliability	Joins, aggregations, window functions, data-quality checks, logging, error handling, and repeatable runs.
Days 61-90	Polish and present the project; apply consistently	Clear repository, data-flow diagram, project walkthrough, interview practice, and tailored applications.